

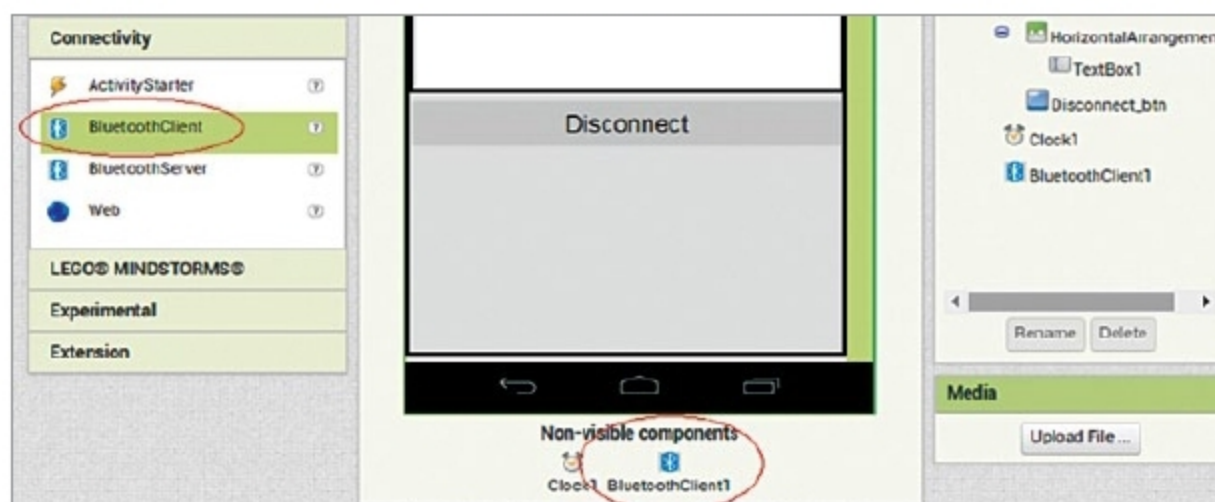


Fig. 13: Adding Bluetooth Client

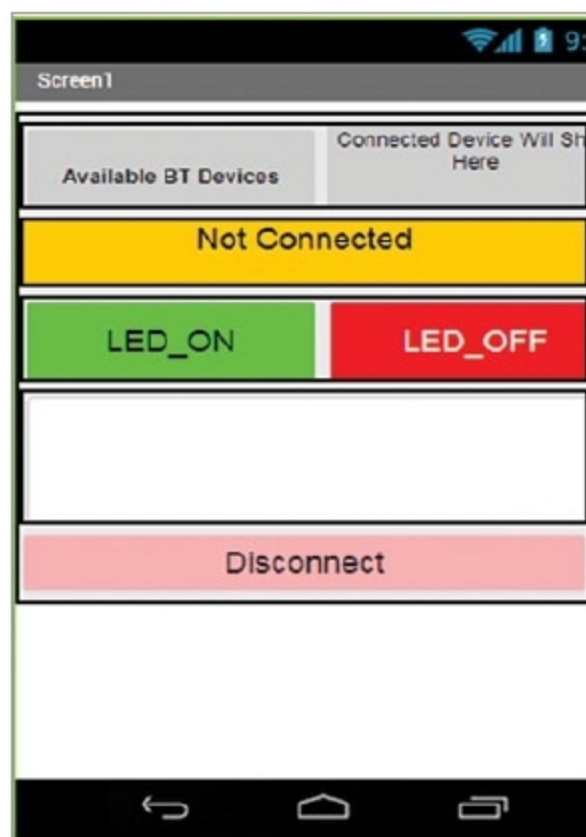


Fig. 14: Final user interface on mobile

which is a hidden component.

Select Palette→User Interface→TextBox and drag it over HorizontalArrangement4. Go to Properties of HorizontalArrangement4 and make the following changes:

- Height: fill parent
- Width: fill parent
- Hint: serial data will show here

Adding disconnect button. Select Palette→User Interface→Button and drag it over VerticalArrangement1. Go to Properties of Button and make the following changes:

- Background colour: pink
- Font size: 19
- Height: automatic
- Width: fill parent
- Text: disconnect
- Text alignment: centre

Non-visible component. Now, add some non-visible components. Palette→Sensor→Clock and drag it to Screen1.



Fig. 15: Blocks

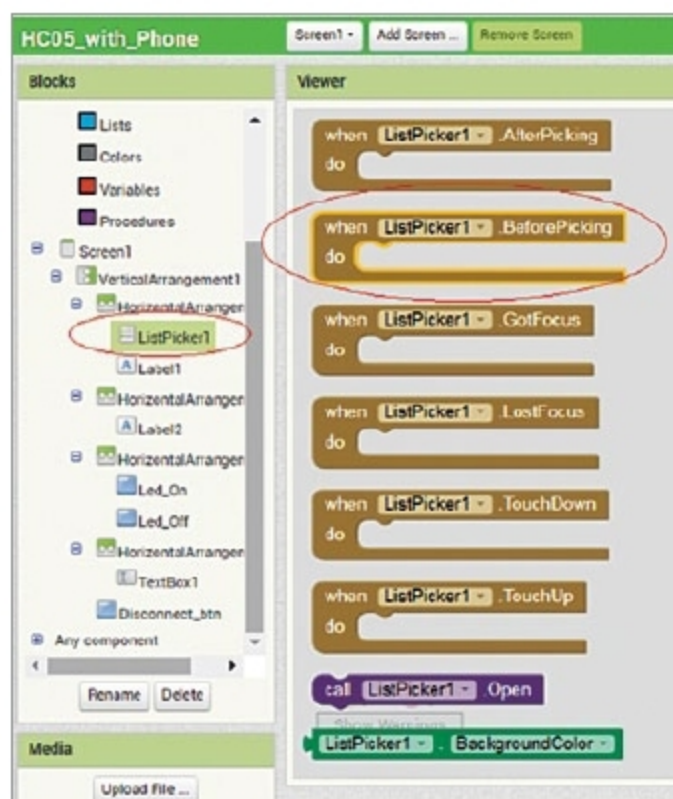


Fig. 16: Viewer window

Adding Bluetooth Client. Bluetooth Client component is used to establish Bluetooth connection. Select Palette→Connectivity→Bluetooth Client and drag it to Screen1.

Finally, the app will look as shown in Fig. 14.

App Inventor Blocks section

Now, let us define the working of each component as per our requirement. Blocks section (Fig. 15) is where you provide the functionality to your selected components. There are built-in blocks that can handle things like Maths, logic and text, which you can use with each component that

you have added. In addition to these components, the project has its own set of blocks specific to its own events, methods and properties.

Click on Blocks at the top-right corner in Screen1. You will get a new window. Click on ListPicker1, as shown in Fig. 16. Select ListPicker1 BeforePicking event, and drag and drop it over Viewer window.

From ListPicker1, select Set ListPicker1.Elements To, and drag and drop it over Viewer window. Then, go to BluetoothClient1 under Screen1 and select BluetoothClient1.AddressAndNames, and drag and drop it over Viewer window. Arrange all events in a similar fashion, as shown in Fig. 17.

BeforePicking(). This is the event to raise when Button is clicked. The event occurs before the list of items is displayed. It is used to prepare the list before it is shown. It sets the elements of ListPicker to available Bluetooth devices by their name and address.

AfterPicking(). This event is to be raised after the picker activity returns its result and properties have been filled in. Select ListPicker1.AfterPicking and go to built-in block Logic. Select another if then event, as shown in Fig. 17.

If if condition is satisfied or true, function BluetoothClient1 will be connected to a device from ListPicker section. It will set label1 text as Connected BT Device and label2 text as Connected, and change the text colour to green.

Now, add blocks for LED_ON and LED_OFF buttons, as shown in Fig. 18. Here, send 1 to turn on the LED and 0 to turn it off. You can also send characters or strings according to the code.

Next, add an event for LED_ON button. When this button is clicked, call BluetoothClient1.SendText function to send text as 1. Do the same for LED_OFF; send 0 on clicking.

Adding clock. Select an event from Clock1.Timer, as shown in Fig. 19. Here, clock is used to receive data from the client.

From Blocks on the left side, select and logic and drag it over Viewer window. This returns true if both conditions are satisfied. Then, select Blue-